



Plan and Code

Quick facilitation notes & answer key

Goal

The unit finale: students plan and write a full program to a goal, meet a harder goal that mixes forward and back, then alter their program for a new goal. This brings together writing/executing code (C3.1) and reading/altering code and describing the effect (C3.2) in one task.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 with a star on 5, and move cards, so students can lay out and test their plan before writing it.

Run it (about 20-25 min)

1. I DO: model Goal 1 — count from 1 to 5 (4 steps forward) and split it: 'forward 2, forward 2.' Run it to land on 5.
2. WE DO: Exercise 1 — students plan their own blocks to reach 5, predict, then run. Exercise 2 (challenge) — from 2 to 3 using one forward and one back (e.g. forward 2, back 1).
3. YOU DO: Exercise 3 — change one block of the Goal 1 code so Bit reaches 4 (e.g. last forward 2 → forward 1).
4. Celebrate: 'You can read, write, fix, and change code — that is what coders do.'

Answer key (modelled by the run-gate)

Ex 1 — start 1 → 5: e.g. forward 2, forward 2 (any two forward blocks that total 4, like forward 1 + forward 3).

Ex 2 — start 2 → 3 with one forward + one back: e.g. forward 2, back 1 (2 → 4 → 3). Forward 1 then back 0 is not allowed; the pair must net +1.

Ex 3 — change the last forward 2 to forward 1: 1 → 3 → 4. Bit lands on 4, one less, because the last step is now one smaller.

Why it works: the start sets the number and the moves run in order; plan the total change (forward adds, back subtracts) to match the distance to the goal.

Watch for & differentiate

Common slip in Ex 2: students use two forward or two back blocks — re-read 'one forward AND one back' and have them net +1 from 2 to 3.

Simplify: allow a single move block for Goal 1 (forward 4) for students who need it.

Extend: ask for a second, different program for Goal 1, or a plan that reaches 5 from start 0.

Success indicator: the child writes a working program for each goal and correctly alters the Goal 1 code to land on 4.